What preposition placement depends on in learner English

A usage-based corpus study

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Preposition placement (PrepPlace) in English RCs

(1) a. a subject [**on** which they had never been able to agree]

b. a subject [which they had never been able to agree **on**]

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Method A sample of relative clauses from English native and nonnative corpora was subjected to a multivariate logistic regression analysis predicting prepPlace from multiple predictor variables.

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Question What determines the variation between preposition fronting and stranding in relative clauses in English as a second language?

Method A sample of relative clauses from English native and nonnative corpora was subjected to a multivariate logistic regression analysis predicting prepPlace from multiple predictor variables.

Results Nonnative prepPlace dependend (a) on different input distributions which changed with proficiency, (b) on cross-linguistic influence of similar native-language structures, and (c) on specific lexical items and strings, among others.

Overview

1. Introduction

2. What prepPlace depends on

3. The current study
3.1 Sample
3.2 Results

4. Discussion

5. Brief outlook

Section 2

What prepPlace depends on

What prepPlace depends on

Frequency distribution
Cross-linguistic similarity
Meaning of the RC filler
Specific lexical items and strings
Complexity of the RC
Meaning of the RC
Register
Modality
...

Gries (2002), Hoffmann (2011), Trotta (2000)

Frequency distribution in English

Table: Distribution of preposition placement by clause type in English (ICE-GB)

	Clause type				
Placement	*Wh*-RCs	Non-*wh*-RCs	...	*Wh*-questions	Total
Fronting	692	0	...	12	708
Stranding	69	419	...	140	882
Total	761	419	...	152	1590

Note. RC = relative clause. Adapted from Hoffmann (2011, pp. 120 & 158).

(Bardovi-Harlig, 1987)

Frequency distribution in English

Table: Frequency distribution of fronting and stranding in *wh*-RCs in different corpora of spoken and written English

	Placement				
Written corpora	Fronting	%	Stranding	%	Total
Bengtsson (1996)	62	95	3	5	65
Van den Eynden (1996)	179	97	6	3	185
Johansson and Geisler (1998) [LOB]	1053	97	27	3	1080
Trotta (2000) [Brown]	1054	99	12	1	1066
Total (written)	2348	98	48	2	2396
Spoken corpora	Fronting	%	Stranding	%	Total
Quirk (1957)	86	83	18	17	104
Johansson and Geisler (1998) [LLC]	190	79	50	21	240
Johansson and Geisler (1998) [BIRM]	799	86	129	14	928
Johansson and Geisler (1998) [BNC]	147	69	65	31	212
Total (spoken)	1222	82	262	18	1484
Total	4262	92	379	8	4641

Cross-linguistic similarity

(2) a. *die Münze, von der ich gesprochen habe*
the coin of which I spoken have

"the coin about which I talked"

(Eisenberg et al., 2009, p. 1031)

b. **die Münze, der ich von gesprochen habe*
the coin which I of spoken have

English	N	PREP *wh* SUBJ V (OBJ)
German	N	PREP *wh* SUBJ (OBJ) V

Cross-linguistic similarity

(3) a. *wǒ yòng jùzi xiūlǐ shuǐ guǎnzi*
I with saw repair water pipe

"I repair the water pipe with the saw."

b. *wǒ xiūlǐ shuǐ guǎnzi de jùzi*
I repair water pipe REL saw

"the saw with which I repair the water pipe"

(Li & Thompson, 1981, p. 582)

English		N	PREP *wh* SUBJ V (OBJ)
German		N	PREP *wh* SUBJ (OBJ) V
Chinese	SUBJ V (OBJ) *de*	N	

(Mazurkewich, 1985)

Specific lexical items and strings

Biber et al. noted that "[f]orms which are typically used as stranded prepositions are those which are closely linked to a preceding word" (1999, p. 106).

(4) a. a subject **on** which they had never been able to agree

b. a subject **on which** they had never been able to agree

c. a subject which they had never been able to **agree on**

(Gries, 2002, Hawkins, 1999, Quirk et al. 1985, p. 664)

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b. a subject **on which** they had never been able to agree

c. a subject which they had never been able to **agree on**

(Gries, 2002, Hawkins, 1999, Quirk et al. 1985, p. 664)

	Item	Frequency (BNC)
on	×	
agree on		×
on which		×

Section 3

The current study

Sample

Corpora	**Learner group**		**Proficiency**	
	Native	Nonnative	Novice	Advanced
ICE-GB	×			×
LCN	×		×	
ICLE		×	×	
YELC		×	×	

Sample

Corpora	Learner group		Proficiency	
	Native	Nonnative	Novice	Advanced
ICE-GB	×			×
LCN	×		×	
ICLE		×	×	
YELC		×	×	

L1	**Corpora**	
	ICLE	YELC
German	×	
French	×	
Spanish	×	
Italian	×	
Chinese	×	
Japanese	×	
Korean		×

Sample

Corpora	Learner group		Proficiency	
	Native	Nonnative	Novice	Advanced
ICE-GB	×			×
LCN	×		×	
ICLE		×	×	
YELC		×	×	

L1	Corpora	
	ICLE	YELC
German	×	
French	×	
Spanish	×	
Italian	×	
Chinese	×	
Japanese	×	
Korean		×

Table: Distribution of fronting and stranding across different corpora

	Fronting	Stranding	Sum
ICE-GB	417	78	495
LCN	211	172	383
ICLE, YELC	853	567	1420
Sum	1481	817	2298

Results

Table: Distribution of fronting and stranding across learner groups

	Fronting	Stranding
Native writers (ICE, LCN)	71.53	28.47
Nonnative writers (ICLE, YELC)	60.07	39.93

Note. $X^2(1) = 30.58, p \leq .001$

Results

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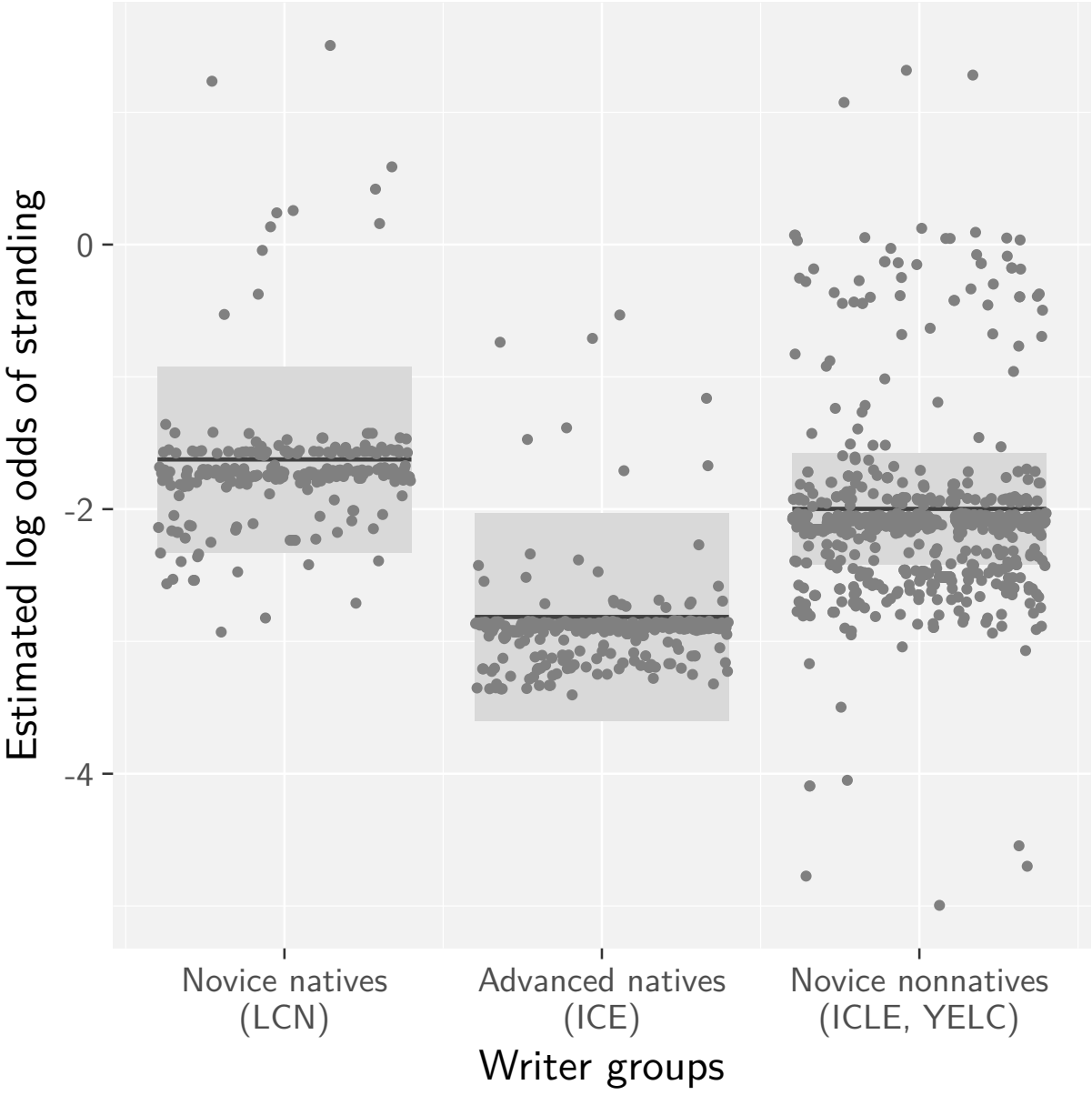
Table: Distribution of fronting and stranding across levels of proficiency

	Fronting	Stranding
Advanced writers (ICE)	84.24	15.76
Novice writers (LCN, ICLE, YELC)	59.01	40.99

Note. $X^2(1) = 106.8, p \leq .001$

Results


Estimated log odds of stranding
0
-2
-4
Novice natives
(LCN)
Advanced natives
(ICE)
Novice nonnatives
(ICLE, YELC)
Writer groups

Results

Table: Output of a binary mixed-effects regression model with random intercepts for prepositions and corpus files; $b^{\,p}$ (SE)

Intercept	-3.45^{***} (0.79)
Participant	1.97^{**} (0.64)
Supplement	-0.79 (0.84)
That	7.95^{***} (1.11)
What/-ever	4.76^{***} (0.72)
Where	1.86 (1.14)
Whom/Who/Whose	-1.70^{**} (0.59)
Advanced native writers (ICE)	$\mathbf{-1.58^{**}}$ **(0.53)**
Novice nonnative writers (ICLE, YELC)	$\mathbf{-0.62}$ **(0.40)**
Frequency preposition	1.23^{**} (0.45)
Frequency preposition-item (e.g. *in which*)	-1.53^{**} (0.56)
Frequency item-preposition (e.g. *agree on*)	1.06^{**} (0.32)
Observations	1,536

Note. $^{*}p \leq .05$; $^{**}p \leq .01$; $^{***}p \leq .001$.

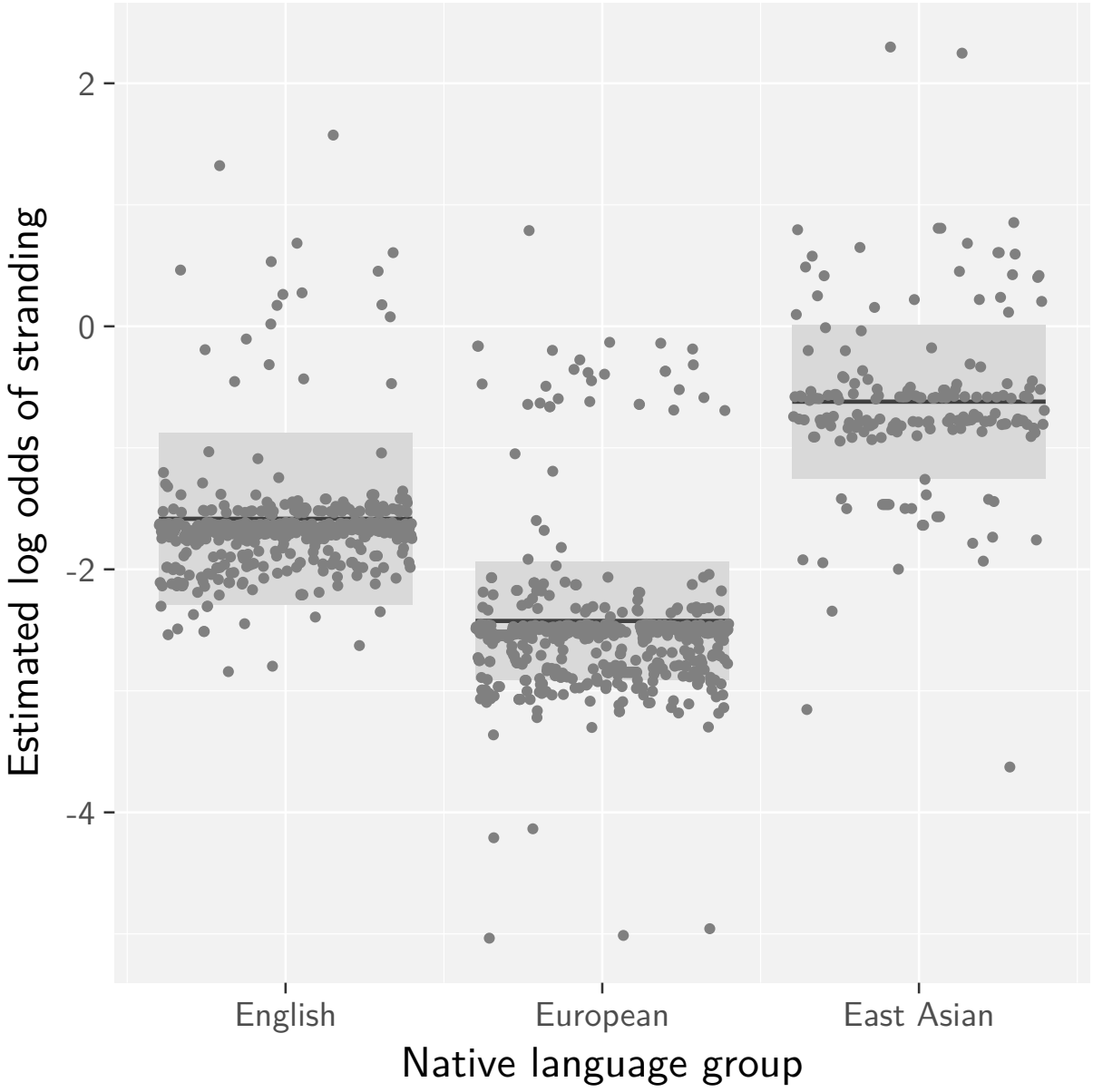
Results

Table: Distribution of fronting and stranding across native language groups

	Fronting	Stranding
English L1	71.53	28.47
European L1 (Ger, Fre, Ita, Spa)	65.79	34.21
East Asian L1 (Chi, Jap, Kor)	40.00	60.00

Note. $X^2(2) = 102.24, p \leq .001$.

Results


2
0
-2
-4
Estimated log odds of stranding
English
European
East Asian
Native language group

Results

Table: Output of a binary mixed-effects regression model with random intercepts for prepositions and corpus files; $b^{\,p}$ (SE)

Intercept	-5.92^{***} (1.60)
European (Ger, Fre, Ita, Spa)	$\mathbf{-1.34^{*}}$ **(0.64)**
East Asian (Chi, Jap, Kor)	**1.10 (0.73)**
Participant	2.27^{**} (0.77)
Supplement	-0.83 (0.99)
That	1.83^{*} (0.77)
What/-ever	9.10^{***} (2.39)
Where	5.72^{***} (1.70)
Whom/Who/Whose	1.67 (1.47)
Novice writers (ICLE, YELC, LNC)	-2.04^{*} (0.96)
Frequency preposition	1.36^{*} (0.61)
Frequency preposition-item (e.g. *in which*)	-1.53^{**} (0.56)
Frequency item-preposition (e.g. *agree on*)	1.06^{**} (0.32)
Observations	1,536

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Results

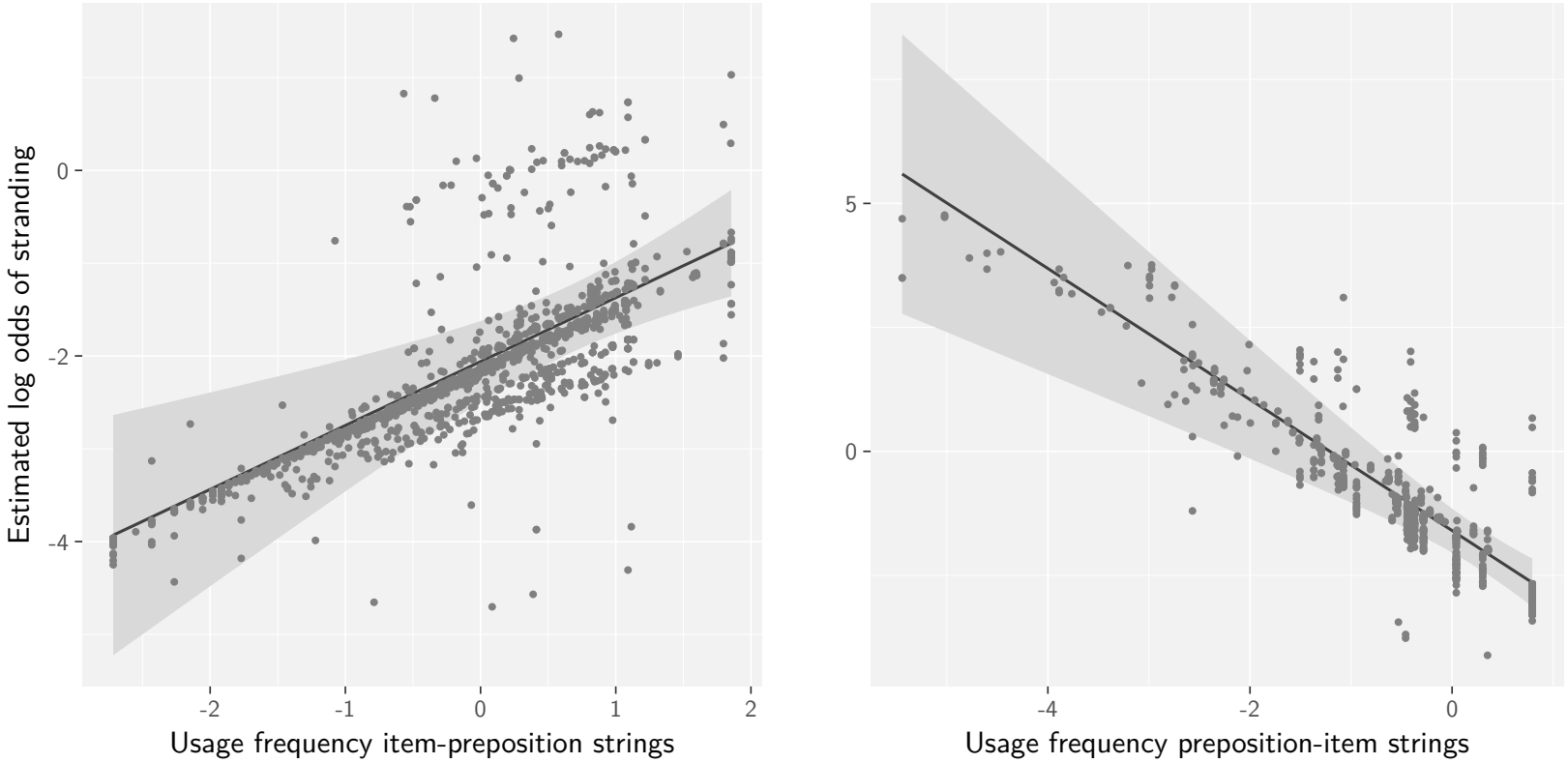

Figure: Estimated effects of string frequencies on preposition placement

Results

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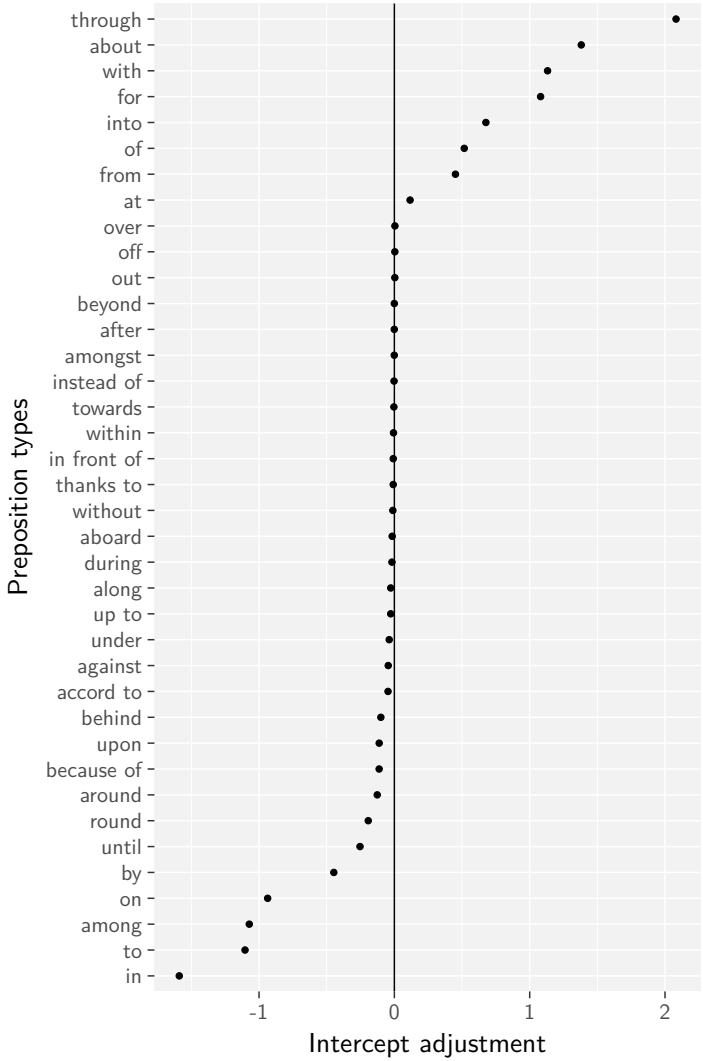
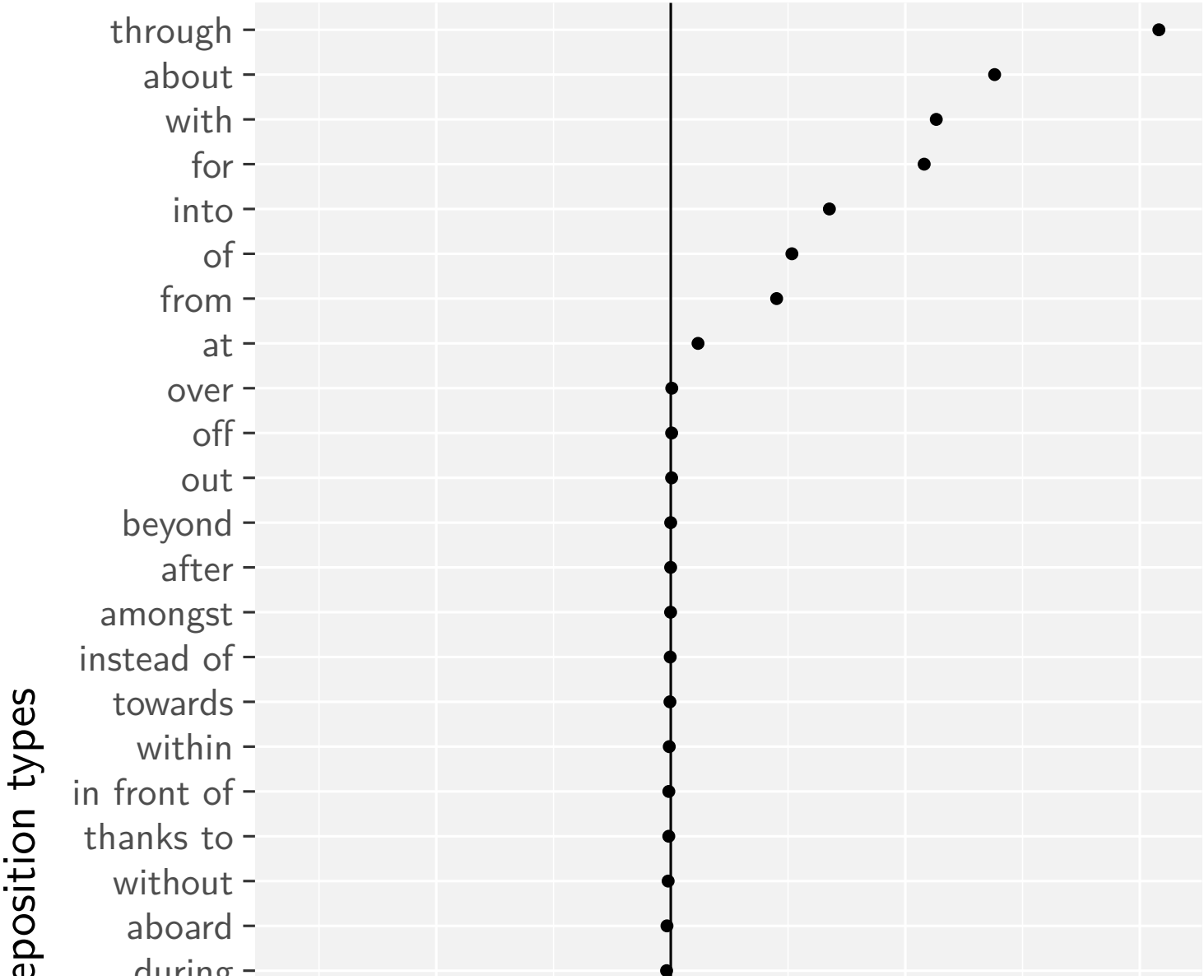


Figure: Intercepts adjustments for preposition types

Results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Results

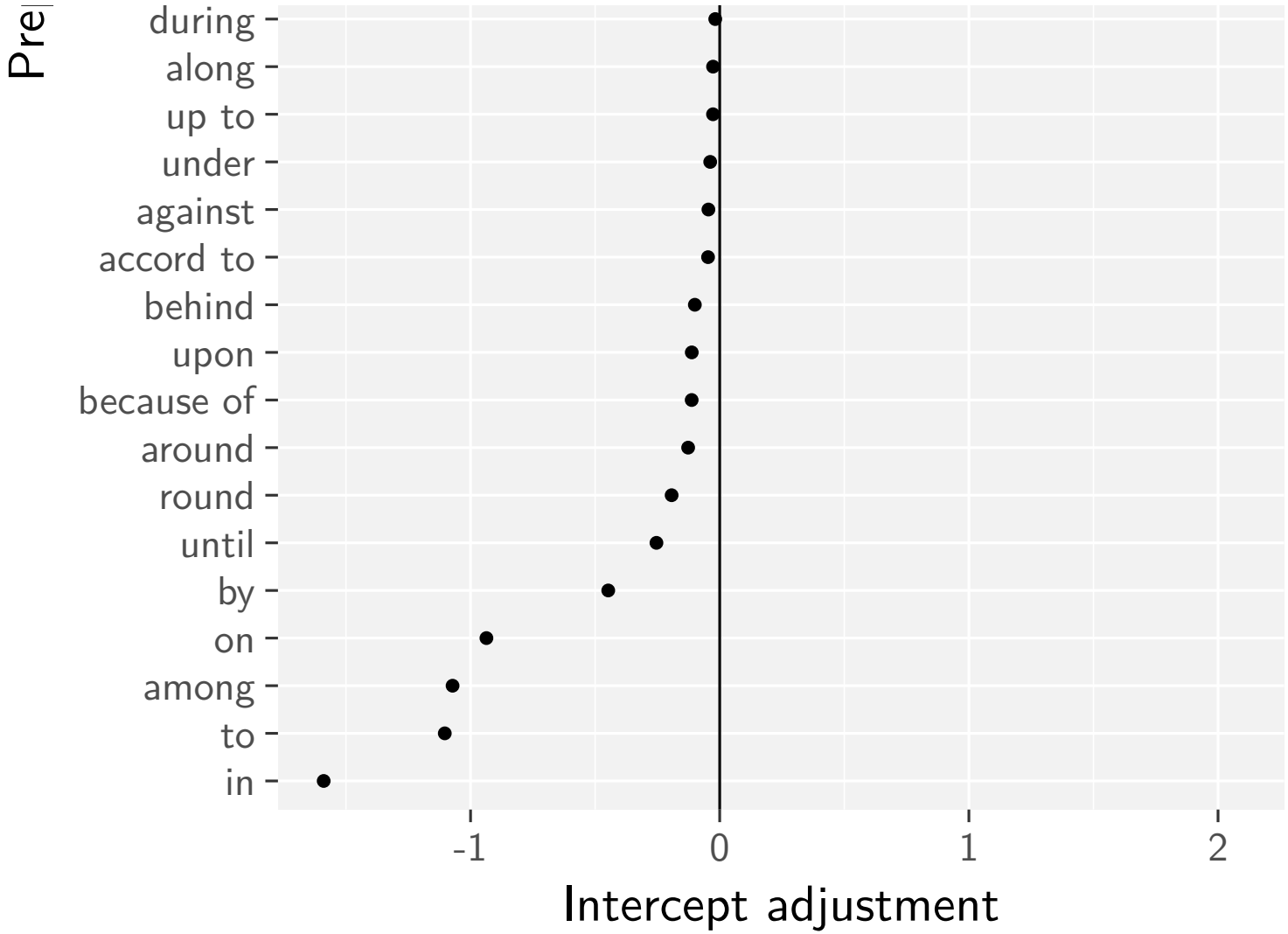

Figure: Intercepts adjustments for preposition

Section 4

Discussion

Sensitivity to usage changes with proficiency

Assumption Language learning is input-driven

(Diessel, 2017)

(Ellis, Simpson-Vlach, & Maynard, 2008; Jach, 2018; Siyanova & Schmitt, 2008)

Sensitivity to usage changes with proficiency

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Results Stranding was associated with novice writers.

- Novice writers were attuned to total relative frequency.
- Advanced writers were attuned to conditional relative frequency.

(Ellis et al., 2008; Jach, 2018; Siyanova & Schmitt, 2008)

Cross-linguistic similarity matters

Assumption Cross-linguistic similarity results in shared representations (Hall, Cheng, & Carlson, 2006; Höder, 2012)

(Hartsuiker & Bernolet, 2017; Hartsuiker, Pickering, & Veltkamp, 2004; Runnqvist, Gollan, Costa, & Ferreira, 2013)

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English		N	PREP *wh* SUBJ V (OBJ)
German		N	PREP *wh* SUBJ (OBJ) V
Chinese	SUBJ V (OBJ) de	N	

(Hartsuiker & Bernolet, 2017; Hartsuiker et al., 2004; Runnqvist et al., 2013)

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English		N	PREP *wh* SUBJ V (OBJ)
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Chinese	SUBJ V (OBJ) de	N	

Results Fronting was associated with European nonnative writers.

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Results Fronting was associated with European nonnative writers.

- Cross-linguistic similarity boosted acquisition of fronting
- ... and resulted in shared fronting representation
- ... for European writers

(Hartsuiker & Bernolet, 2017; Hartsuiker et al., 2004; Runnqvist et al., 2013)

What fronting and stranding constructions look like

Assumption Constructions are item-specific and cross-linked (Diessel, 2016, 2018; Ellis, Römer, & O'Donnell, 2016)

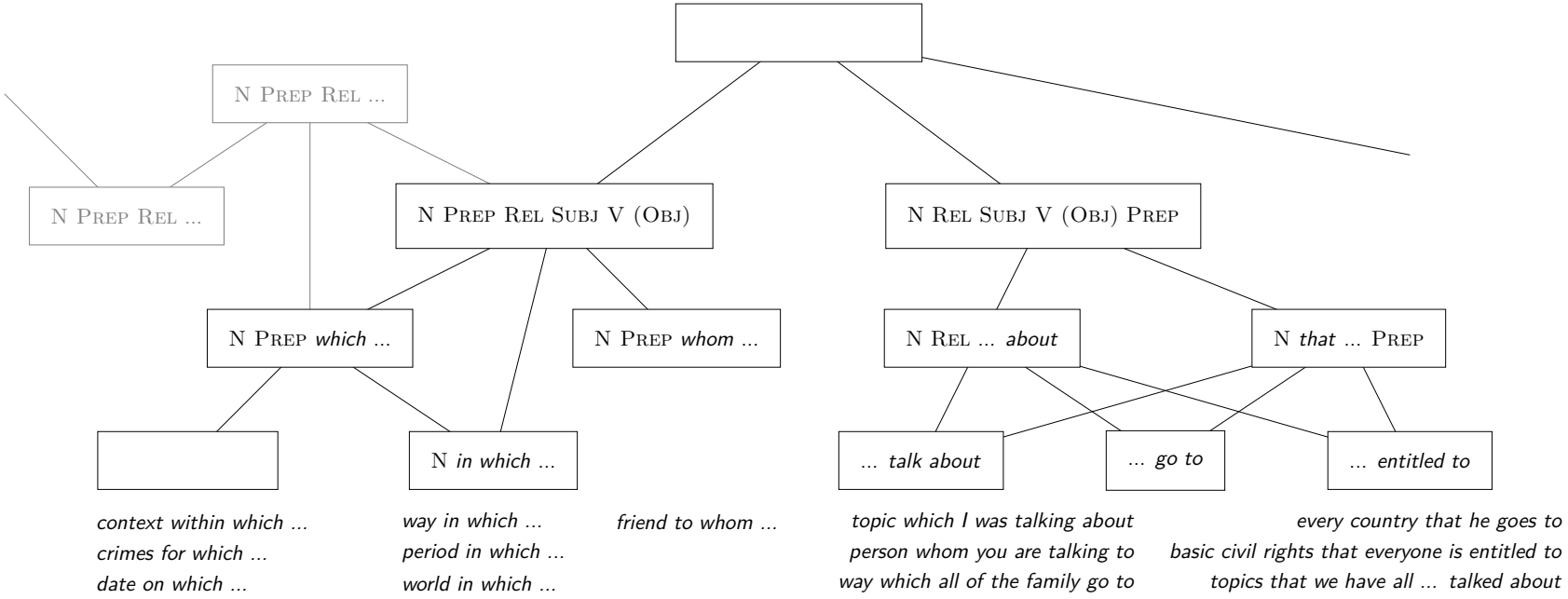


Figure: Network of fronting and stranding RC constructions

What fronting and stranding constructions look like

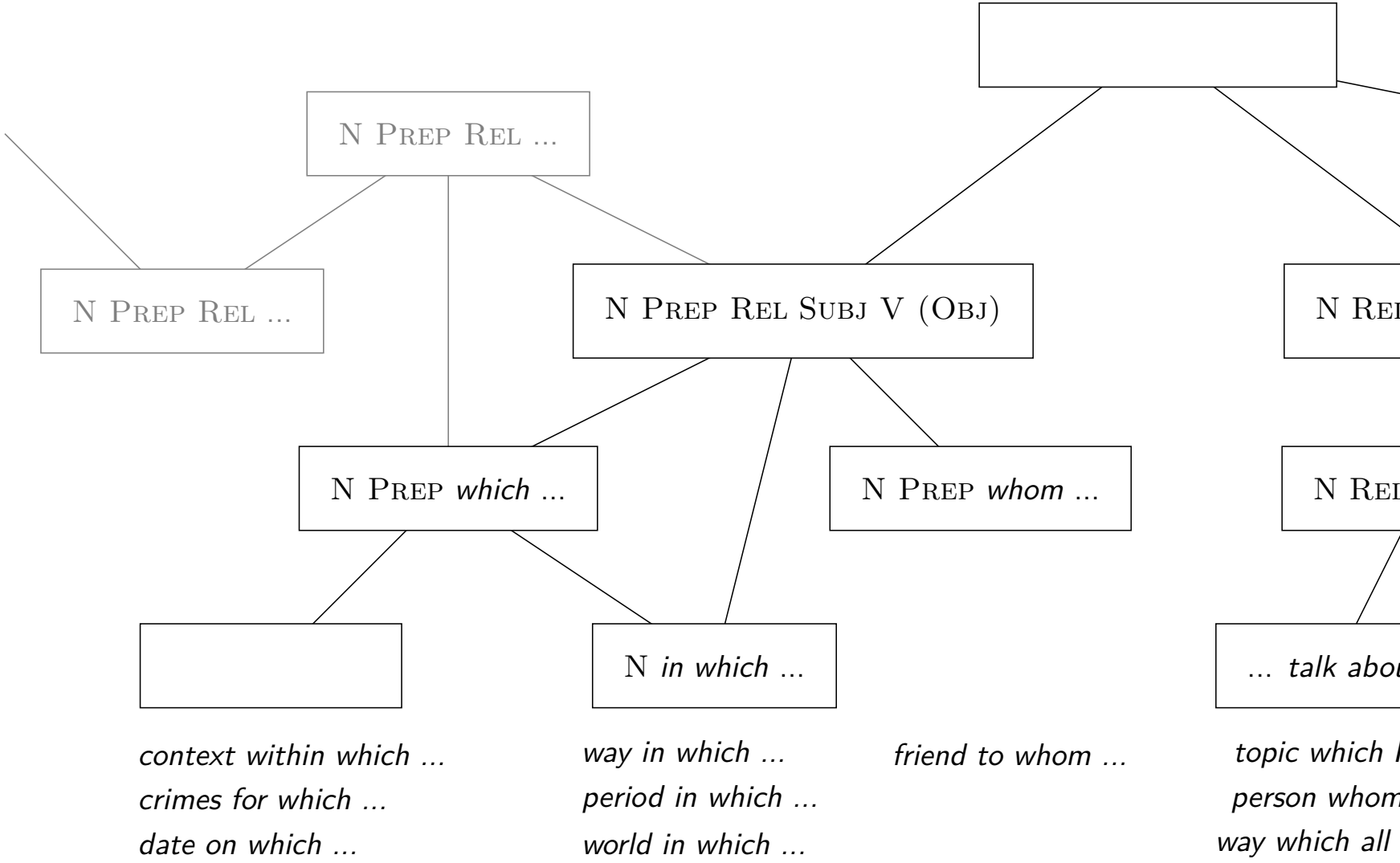
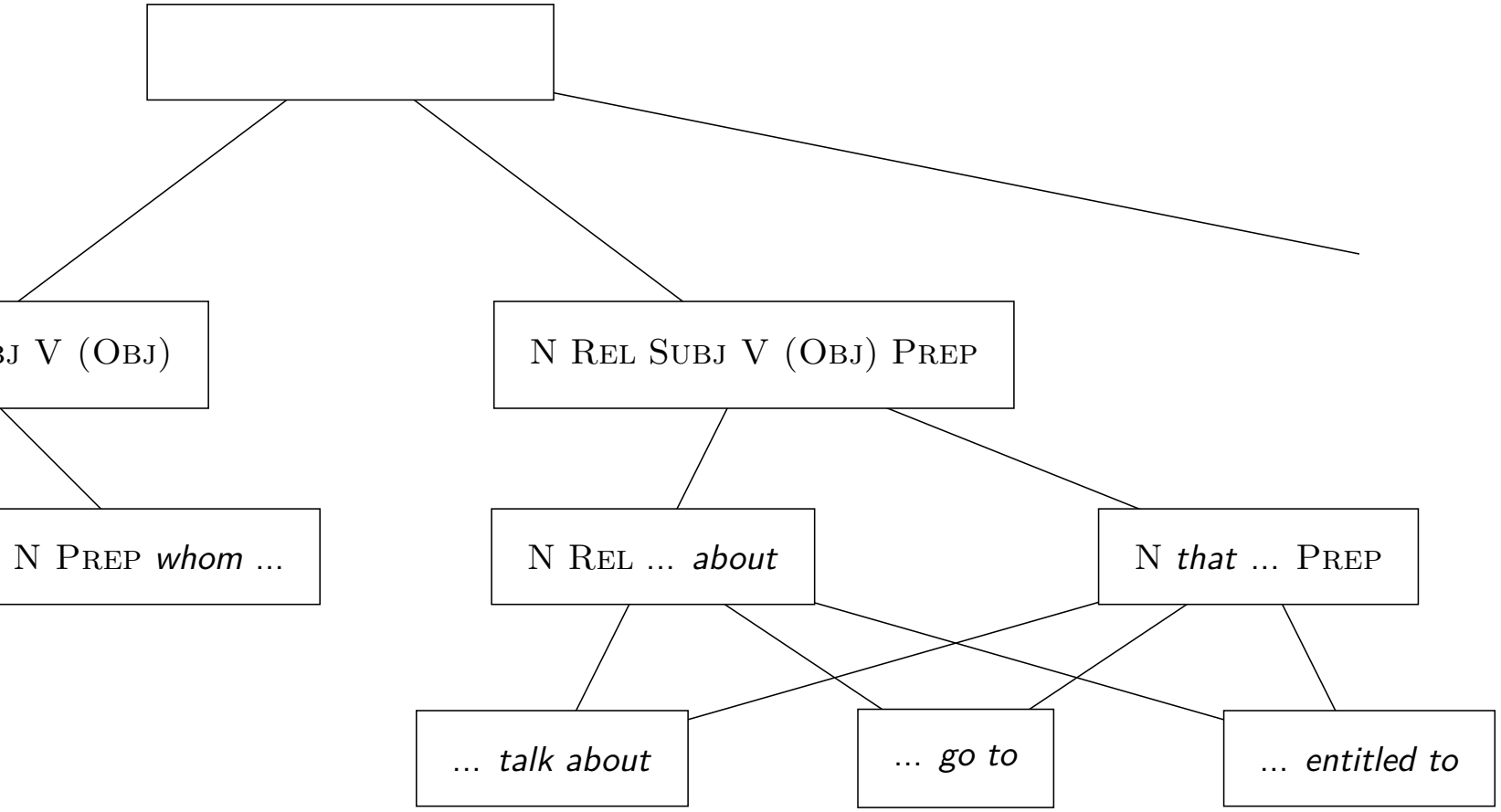


Figure: Network of fronting and stranding RC co

What fronting and stranding constructions look like



friend to whom ...

topic which I was talking about
person whom you are talking to
way which all of the family go to

every country that he goes to
basic civil rights that everyone is entitled to
topics that we have all ... talked about

nd stranding RC constructions

Section 5

Brief outlook

Information structure

"A [noun phrase] can only be extracted out of a [preposition phrase] which may be interpreted as being more important (newer) than the rest of the sentence." (Takami, 1992, p. 43)

(5) a. Which party did John write **the** letter after?

b. (?)Which party did John write **a** letter after?

c. ??Which party did John write **some** letters after?

d. *Which park did you find **more/any** rabbits in?

Takami (1988, 1992)

Thank you for your attention.

References I

- Bardovi-Harlig, K. (1987). Markedness and salience in second-language acquisition. *Language Learning*, *3*, 385–407. doi:10.1111/j.1467-1770.1987.tb00577.x
- Bengtsson, P. (1996). *Prepositional relatives in English: A diachronic case study of three authors*. Department of English, Göteborg University.
- Biber, D., Johansson, S., Leech, G., Conrad, S., & Finegan, E. (1999). *Longman grammar of written and spoken English*. London, England: Longman.
- Diessel, H. (2016). Frequency and lexical specificity in grammar: A critical review. In H. Behrens & S. Pfänder (Eds.), *Experience counts: Frequency effects in language* (pp. 209–238). Berlin, Germany: De Gruyter.
- Diessel, H. (2017). Usage-based linguistics. Retrieved May 17, 2018, from https://doi.org/10.1093/acrefore/9780199384655.013.363.
- Diessel, H. (2018). *The grammar network*. Manuscript submitted for publication.

References II

- Eisenberg, P., Peters, J., Gallmann, P., Fabricius-Hansen, C., Nübling, D., Barz, I., ... Henning, M. (2009). *Duden: Die Grammatik: Unentbehrlich für für richtiges Deutsch* (8th ed.). Mannheim, Germany: Dudenverlag.
- Ellis, N. C., Römer, U., & O'Donnell, M. B. (2016). *Usage-based approaches to language acquisition and processing: cognitive and corpus investigations of construction grammar*. Wiley-Blackwell.
- Ellis, N. C., Simpson-Vlach, R., & Maynard, C. (2008). Formulaic language in native and second language speakers: psycholinguistic, corpus linguistics, and TESOL. *TESOL Quarterly*, *42*(3), 375–396.
- Gries, S. T. (2002). Preposition stranding in English: Predicting speakers' behaviour. In V. Samiian (Ed.), *Proceedings of the Western Conference on Linguistics* (Vol. 12, pp. 230–41). Fresno, CA: Department of Linguistics at California State University.
- Hall, J. K., Cheng, A., & Carlson, M. T. (2006). Reconceptualizing multicompetence as a theory of language knowledge. *Applied Linguistics*, *27*(2), 220–240.

References III

- Hartsuiker, R. J. & Bernolet, S. (2017). The development of shared syntax in second language learning. *Bilingualism: Language and Cognition*, *20*(2), 219–234. doi:10.1017/S1366728915000164
- Hartsuiker, R. J., Pickering, M. J., & Veltkamp, E. (2004). Is syntax separate or shared between languages? Cross-linguistic syntactic priming in spanish-english bilinguals. *Psychological Science*, *15*(6), 409–414. doi:10.1111/j.0956-7976.2004.00693.x
- Hawkins, J. A. (1999). Processing complexity and filler-gap dependencies across grammars. *Language*, *75*(2), 244–285.
- Höder, S. (2012). Multilingual constructions: A diasystematic approach to common structures. In K. Braunmüller & C. Gabriel (Eds.), *Multilingual individuals and multilingual societies* (pp. 241–258). Amsterdam, The Netherlands: Benjamins. doi:10.1075/hsm.13.17hod
- Hoffmann, T. (2011). *Preposition placement in English: A usage-based approach*. Cambridge: Cambridge University Press. doi:10.1017/CBO9780511933868

References IV

- Jach, D. (2018). A usage-based approach to preposition placement in English as a second language. *Language Learning*, *68*(1), 271–304. doi:10.1111/lang.12277
- Johansson, C. & Geisler, C. (1998). Pied piping in spoken English. *Language and Computers*, *23*, 67–82.
- Li, C. N. & Thompson, S. A. (1981). *Mandarin Chinese: A functional reference grammar*. Berkeley: University of California Press.
- Mazurkewich, I. (1985). Syntactic markedness and language acquisition. *Studies in Second Language Acquisition*, *7*(1), 15–35. doi:10.1017/S0272263100005131
- Quirk, R. (1957, January). Relative clauses in educated spoken English. *English Studies*, *38*(1-6), 97–109. doi:10.1080/00138385708596993
- Quirk, R., Greenbaum, S., Leech, G., & Svartvik, J. (1985). *A comprehensive grammar of the English language*. London, England: Longman.

References V

- Runnqvist, E., Gollan, T. H., Costa, A., & Ferreira, V. S. (2013). A disadvantage in bilingual sentence production modulated by syntactic frequency and similarity across languages. *Cognition*, *129*(2), 256–263. doi:10.1016/j.cognition.2013.07.008
- Siyanova, A. & Schmitt, N. (2008). L2 learner production and processing of collocation: A multi-study perspective. *The Canadian Modern Language Review*, *64*(3), 429–458.
- Takami, K. (1988). Preposition stranding: Arguments against syntactic analyses and an alternative functional explanation. *Lingua*, *76*(4), 299–335. doi:10.1016/0024-3841(88)90022-8
- Takami, K. (1992). *Preposition stranding: From syntactic to functional analyses*. Berlin, Germany: De Gruyter.
- Trotta, J. (2000). *Wh-clauses in English: Aspects of theory and description*. Amsterdam, The Netherlands: Rodopi.
- Van den Eynden, N. (1996). Aspects of preposition placement in English. In J. Klemola, M. Kytö, & M. Rissanen (Eds.), *Speech past and present: Studies in English dialectology in memory of Ossi Ihalainen* (pp. 426–446). Frankfurt am Main, Germany: Peter Lang.